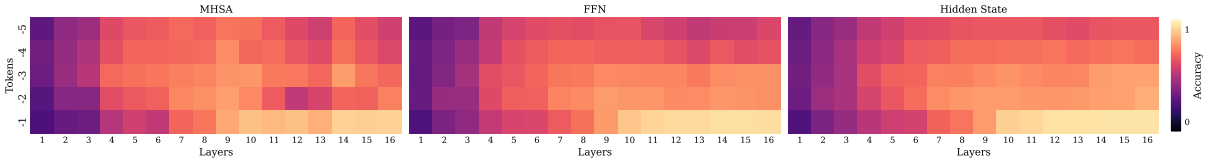

Figure 18: Probing test accuracy on different tokens of Llama 3.2 1B across all layers. We observe a consistent increase in signal strength from earlier to later tokens and from lower to higher layers.

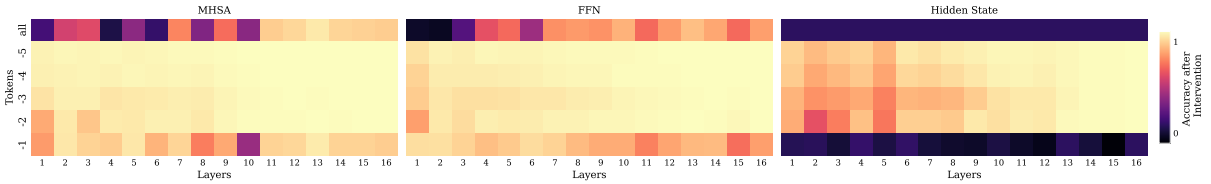

Figure 19: Zero intervention accuracy on different token indices of llama 3.2 1B for all layers with span = 1. The vertical dimension shows tokens and *"all"* means to knock out all activations of the specific layers at the same time. MHSA units beyond the critical layer ($l = 10$) have minimal impact on the causal path, even when an entire layer is knocked out.

(i.e., when all units in a layer are knocked out). Notably, the results indicate that MSHA units beyond the critical layer $l = 10$ contribute minimally to the causal path, even when an entire layer is knocked out.

C.4 Control Experiment on an Isomorphic Task

Here, we investigate whether the observed outcomes could be attributed solely to syntactic features and task structure rather than the target task of emotion processing. To assess this, we conduct an isomorphic experiment in which we modify the task to focus purely on syntax—predicting the first word in the sequence—and repeat the activation patching procedure as described in Section 6. The altered prompt is shown below:

> **What is the first word in the following contexts?** Context: My dog died last week. Answer: **My**; Context: I saw moldy food. Answer: **I**; Context: I could see my friend after a long time. Answer:

As shown in Figures 22 and 23, MHSA units in the final layers of the model are most critical for this syntactic task, which contrasts significantly with the emotion patching findings. Additionally, we observe weaker evidence of functional localization based on the patching results from the isomorphic control experiment.

C.5 Robustness to Prompt Design

To evaluate the model's robustness to various prompts, we designed a variety of prompt templates, as illustrated in Table 2. Figure 24 demonstrates the distribution of the next word emotion label predictions for the different prompt templates and different numbers of few-shot examples. Figure 25 demonstrates the final accuracy of these tests. Noteworthy that again in the following interpretability experiments, we confine our focus only on the samples that the model could predict correctly using each specific prompt.

Figures 26 and 27 validate that the probing and patching results presented earlier in the paper are robust to various prompt templates. For probing, we see that the model has determined the predicted label by the mid layers of the model, a result that is consistent across various prompt templates and the number of few shot examples provided. For activation patching, we also get consistent results across various prompt templates and with varied numbers of few-shot examples.

C.6 Investigating the Incorrectly Classified Samples

As mentioned earlier, we only used the dataset instances on which the LLM could predict the self-reported emotion label with no mistakes. We initially adopted this approach to reduce noise in the dataset, given the subjective nature of emotion labeling. Emotion classification can become a very subjective task in some contexts. For example, in many cases, it is hard to distinguish anger from

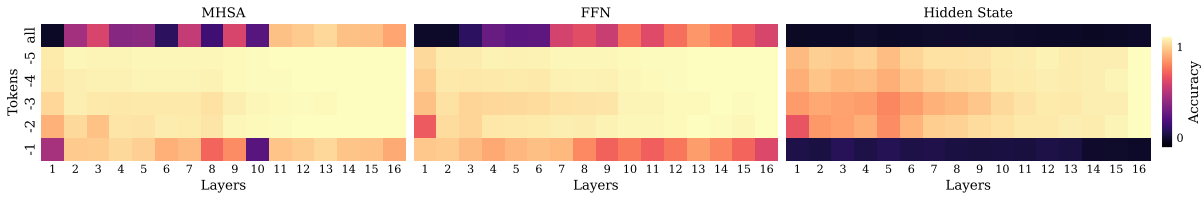

Figure 20: Random intervention accuracy on different token indices of Llama 3.2 1B across all layers with span = 1. The vertical axis represents token positions, where *"all"* denotes the simultaneous knockout of all activations in the specified layers. Notably, MHSA units beyond the critical layer ($l = 10$) contribute minimally to the causal path, even when an entire layer is deactivated.

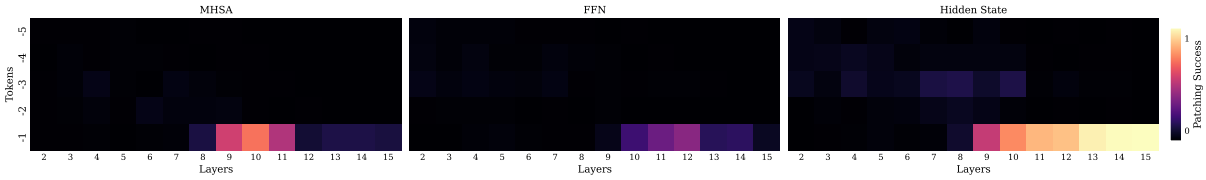

Figure 21: Results of activation patching success on different tokens of Llama 3.2 1B across all layers with span = 3. Clear functional localization is observed in layers 9–11.

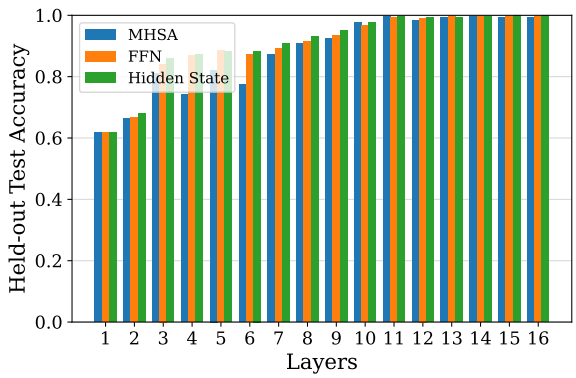

Figure 22: Probing test accuracy of the control isomorphic experiment on the last token of Llama 3.2 1B across all layers.

#	Prompt Template
1	What are the inferred emotions in the following contexts? Context: My first child was born. Answer: joy Context: My dog died last week. Answer: sadness Context: [Input] Answer:
2	Consider this list of emotions: anger, boredom, disgust, fear, guilt, joy, pride, relief, sadness, shame, surprise, trust, neutral. What are the inferred emotions in the following contexts? Context: My first child was born. Answer: joy Context: My dog died last week. Answer: sadness Context: [Input] Answer:
3	Context: My first child was born. Answer: joy Context: My dog died last week. Answer: sadness Context: [Input] Answer:
4	Guess the emotion. Context: My first child was born. Answer: joy Context: My dog died last week. Answer: sadness Context: [Input] Answer:

Table 2: The prompt templates used for experimenting with the language models. The [Input] would be replaced with the sample sentence from the dataset that we are trying to label.

guilt, given the short description provided by the participants. Please see Figure 9, where we show how closely related emotions might be confused. We emphasize that focusing only on correctly classified examples may introduce bias, particularly a bias toward unambiguous emotional content. This is common in much of the prior work in mechanistic interpretability, which focuses on tasks that LLMs perform well on to ensure reliable interpretation.

That being said, analyzing misclassified examples could offer a deeper understanding of model mechanisms. In this section, we conduct a preliminary analysis comparing appraisal patterns between correctly and incorrectly classified cases for each emotion. Interestingly, we found that LLMs tend to exhibit distinct appraisal profiles in misclassified cases, suggesting shifts in the underlying reasoning process.

Table 3 summarizes these findings for three emotions that showed high confusion, presenting the mean and standard deviation of key appraisal dimensions using the appraisal probes we trained, along with t-test comparisons. These results suggest that misclassifications are not simply random errors but may stem from meaningful shifts in how

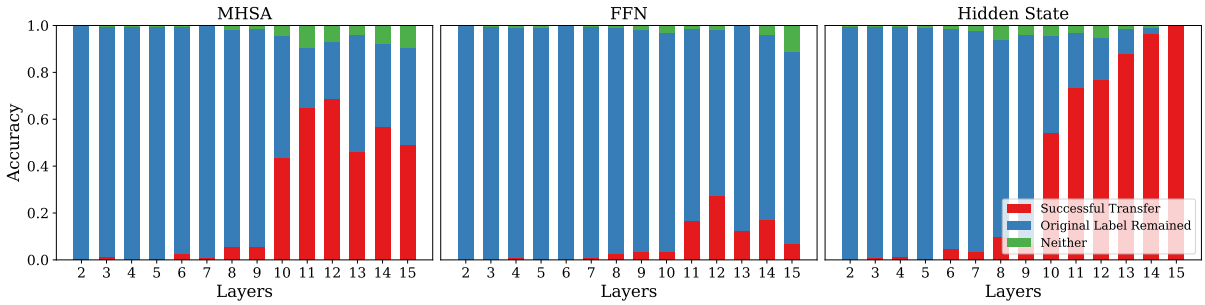

Figure 23: Control isomorphic experiment results for Llama 3.2 1B activation patching across different layers at MHSA, FFN, and hidden states with span = 3, evaluated over 200 source-target pairs. Localization is less evident, with the highest patching performance observed around the final layers.

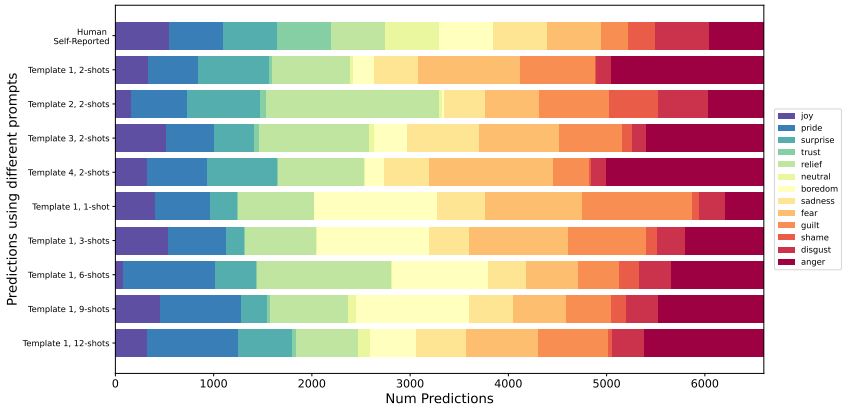

Figure 24: The distribution of next word emotion label predictions for different prompt templates and varied numbers of few shot examples.

the model evaluates appraisal cues.

Emotion	Pleasantness (Correct / Miss)	Others agency (Correct / Miss)
Sadness	$1.3 \pm 0.5/1.6 \pm 0.7$	$2.2 \pm 0.7/3.2 \pm 0.9$
Joy	$4.5 \pm 0.5/4.2 \pm 0.7$	$3.5 \pm 0.7/3.1 \pm 0.8$
Guilt	$1.9 \pm 0.6/1.6 \pm 0.8$	$2.8 \pm 0.7/3.3 \pm 0.7$

Table 3: Comparing the average appraisal values the LLM perceived for correctly classified vs misclassified samples with different emotion labels. All values reported here have p-value less than 0.001

D Direct Emotion Promotion

In Section 9, we showed that it is possible to change the model's output by manipulating appraisal concepts and directing it toward emotions with certain specifications of appraisals. In this section, we show that one can also directly inject a desired specific emotion label output by linearly adding the corresponding emotion vector to the hidden state of the model. More formally, recall the weight vector $\mathbf{w}_e$ for emotion e as introduced in Section 8. We define the emotion promotion modification as

$$\mathbf{x} \leftarrow \mathbf{x} + \beta \frac{\mathbf{w}_e}{\|\mathbf{w}_e\|_2},$$

where $\mathbf{x}$ on the RHS is the activation from the original model at any desired layer or location, and β is the scaling factor that controls the strength of emotion promotion.

Figure 28 shows the results of direct emotion promotion when performed on the hidden states across different layers of Llama 3.2 1B for different target emotion labels. As the figure suggests, direct emotion promotion is not effective when applied to the early layers, which completely aligns with our prior results. However, after layer 9, the success chance greatly improves, especially for large enough values of β. Again, this is a validation of our previous findings which shows that emotion concepts are linearly accessible and modifiable after the mid-layers. But before these layers, even a direct modification may fail since it will be overwritten later by the subsequent layers.